Lesson 42: Contraction and Value Iteration

AI/ML Mathematics

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1 Motivation and Overview

Lesson 41 established that the optimal value function V^* is the unique fixed point of the Bellman optimality operator $\mathcal{T}^*$, and that the Bellman expectation operator $\mathcal{T}^\pi$ is a γ-contraction with fixed point V^π. These results are existential: they tell us that unique solutions exist, but they do not yet provide a concrete, iterative algorithm for computing them.

Suppose we have a large warehouse MDP with thousands of states and want to find the optimal policy. We cannot enumerate all A^S deterministic policies and evaluate each one. Instead, we need an algorithm that starts from an arbitrary initial guess $v^{(0)}$ and iteratively improves it toward V^*. How fast does this process converge? How do we know when to stop? And can we extract a near-optimal policy from an approximate value function?

The tools developed in this lesson – value iteration as a direct application of the Banach fixed-point theorem, its geometric convergence rate, and the connection between approximate fixed points and approximate optimality – answer these questions and form the computational backbone of model-based reinforcement learning.

Why Value Iteration Matters for RL/ML

- **Classical planning**: Value iteration is the primary algorithm for solving known MDPs; it is the standard benchmark for model-based planning and appears in every RL textbook.
- **Q-learning**: The Q-learning update $Q(s,a) \leftarrow (1-\alpha)Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a')]$ is a stochastic, asynchronous version of value iteration applied to the action-value Bellman operator.
- **Fitted value iteration**: Deep RL methods such as DQN perform approximate value iteration by fitting a neural network to the target $r + \gamma \max_{a'} Q(s',a';\theta^-)$, directly implementing the Bellman backup with function approximation.
- **Convergence analysis**: The contraction rate γ derived here provides the template for convergence proofs across RL, from tabular methods to linear function approximation.

Prerequisites. Bellman expectation operator $\mathcal{T}^\pi$, Bellman optimality operator $\mathcal{T}^*$, their contraction properties, the policy improvement theorem, greedy policies (Lesson 41). The Banach fixed-point theorem on complete metric spaces (Lesson 4, Lesson 15). Supremum norm on $\mathbb{R}^S$ (Lesson 5). Geometric series convergence (Lesson 1).

Learning objectives. After completing this lesson, the student will be able to:

- *State* the value iteration algorithm for both V-values and Q-values.
- *Prove* that value iteration converges geometrically to V^* at rate γ using the Banach fixed-point theorem.
- *Derive* the a priori and a posteriori error bounds for value iteration.
- *Compute* the number of iterations required to achieve ε-optimality.
- *Prove* that a policy greedy with respect to an ε-approximate value function is $\frac{2\gamma\varepsilon}{1-\gamma}$-optimal.
- *Apply* value iteration to small MDPs and verify convergence.

2 References

- Puterman [2014, Chapter 6, Sections 6.1–6.3] – Definitive treatment of value iteration, convergence analysis, and error bounds for discounted MDPs.
- Bertsekas [2012, Chapter 1, Sections 1.2–1.3] – Value iteration as contraction iteration, convergence rates, and approximate dynamic programming.
- Denardo [1967] – Unified contraction framework placing value iteration in the broader context of monotone operator theory.
- Sutton and Barto [2018, Chapter 4] – Accessible presentation of value iteration with algorithmic pseudocode and examples.
- Blackwell [1962] – Original proof of the contraction property and existence of optimal policies via iterative methods.

3 The Value Iteration Algorithm

We work throughout with a finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with $|\mathcal{S}| = S$, $|\mathcal{A}| = A$, $R_{\max} = \max_{s,a} |R(s,a)|$, and $\gamma \in [0,1)$. Value functions are elements of $\mathbb{R}^S$ equipped with the supremum norm $\|v\|_\infty = \max_{s\in\mathcal{S}} |v(s)|$.

Recall from Lesson 41 that the Bellman optimality operator $\mathcal{T}^* : \mathbb{R}^S \to \mathbb{R}^S$ is defined by

$$(\mathcal{T}^* v)(s) = \max_{a\in\mathcal{A}} \left[R(s,a) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s,a)\, v(s') \right],$$

and is a γ-contraction on $(\mathbb{R}^S, \|\cdot\|_\infty)$ with unique fixed point V^* (Lesson 41, Theorems 5.2 and 5.3).

Definition 3.1 (Value Iteration)**.** The *value iteration* (VI) algorithm generates a sequence of value functions $\{v^{(n)}\}_{n\geq 0}$ by:

1. **Initialize:** Choose any $v^{(0)} \in \mathbb{R}^S$ (commonly $v^{(0)} = 0$).
2. **Iterate:** For $n = 0, 1, 2, \ldots$, compute

$$v^{(n+1)} = \mathcal{T}^* v^{(n)},$$

   i.e., for each $s \in \mathcal{S}$:

$$v^{(n+1)}(s) = \max_{a\in\mathcal{A}} \left[R(s,a) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s,a)\, v^{(n)}(s') \right].$$

3. **Extract policy:** After termination, return the greedy policy $d^{(n)}(s) = \arg\max_{a\in\mathcal{A}} [R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, v^{(n)}(s')]$.

Remark 3.2. Each iteration of value iteration requires $O(S^2 A)$ arithmetic operations: for each of the S states and A actions, computing the one-step backup $R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) v^{(n)}(s')$ costs $O(S)$ operations. Storage is $O(S)$ for the current iterate.

4 Convergence of Value Iteration

The convergence of value iteration follows directly from the Banach fixed-point theorem applied to the contraction $\mathcal{T}^*$.

Theorem 4.1 (Value Iteration Converges Geometrically)**.** *For any initial $v^{(0)} \in \mathbb{R}^S$, the value iteration sequence $v^{(n+1)} = \mathcal{T}^* v^{(n)}$ satisfies:*

1. $\|v^{(n)} - V^*\|_\infty \le \gamma^n \|v^{(0)} - V^*\|_\infty$ *for all* $n \ge 0$.
2. $v^{(n)} \to V^*$ *in the supremum norm as* $n \to \infty$.
3. *The convergence rate is geometric with ratio* γ.

Proof. Since $\mathcal{T}^*$ is a γ-contraction on the complete metric space $(\mathbb{R}^S, \|\cdot\|_\infty)$ (Lesson 41, Theorem 5.2) with fixed point V^*:

$$\|v^{(n)} - V^*\|_\infty = \|(\mathcal{T}^*)^n v^{(0)} - (\mathcal{T}^*)^n V^*\|_\infty \le \gamma^n \|v^{(0)} - V^*\|_\infty,$$

where we used the contraction property inductively: $\|\mathcal{T}^* u - \mathcal{T}^* w\|_\infty \le \gamma \|u - w\|_\infty$ implies $\|(\mathcal{T}^*)^n u - (\mathcal{T}^*)^n w\|_\infty \le \gamma^n \|u - w\|_\infty$.

Since $\gamma \in [0, 1)$, $\gamma^n \to 0$, giving $v^{(n)} \to V^*$. □

A Priori Error Bound for Value Iteration

Starting from $v^{(0)} = 0$, the value iteration error satisfies

$$\|v^{(n)} - V^*\|_\infty \le \gamma^n \|V^*\|_\infty \le \frac{\gamma^n R_{\max}}{1 - \gamma}.$$

The second inequality uses $\|V^*\|_\infty \le R_{\max}/(1-\gamma)$, which follows from $|G_t| \le R_{\max}/(1-\gamma)$ (Lesson 39, Proposition 5.1).

To achieve $\|v^{(n)} - V^*\|_\infty \le \varepsilon$, it suffices to take

$$n \ge \frac{1}{1-\gamma} \log\left(\frac{R_{\max}}{\varepsilon(1-\gamma)}\right) = \frac{1}{\log(1/\gamma)} \log\left(\frac{R_{\max}}{\varepsilon(1-\gamma)}\right).$$

Example 4.2 (Iteration Count for Common Parameters)**.** For an MDP with $R_{\max} = 1, \gamma = 0.99$, and desired accuracy $\varepsilon = 0.01$:

$$n \ge \frac{1}{\log(1/0.99)} \log\left(\frac{1}{0.01 \cdot 0.01}\right) = \frac{\log(10000)}{\log(1/0.99)} \approx \frac{9.21}{0.01005} \approx 917.$$

With $\gamma = 0.9$ instead: $n \ge \log(1000)/\log(1/0.9) \approx 6.91/0.1054 \approx 66$. The iteration count is highly sensitive to γ: closer to 1 means much slower convergence.

Dependence on $1/(1-\gamma)$

The iteration complexity of value iteration scales as $O(\frac{1}{1-\gamma} \log \frac{1}{\varepsilon(1-\gamma)})$. For γ close to 1, the factor $1/(1-\gamma)$ can be very large (e.g., $1/(1-0.999) = 1000$). This makes value iteration impractical for near-undiscounted problems. Policy iteration, which converges in at most A^S steps independent of γ, can be preferable in such cases.

We now establish a more practical error bound that uses only quantities computable during the algorithm, rather than the unknown V^*.

Theorem 4.3 (A Posteriori Error Bound)**.** *For the value iteration sequence $\{v^{(n)}\}$,*

$$\|v^{(n)} - V^*\|_\infty \le \frac{\gamma}{1-\gamma} \|v^{(n)} - v^{(n-1)}\|_\infty$$

for all $n \ge 1$.

Proof. By the triangle inequality and the contraction property:

$$\|v^{(n)} - V^*\|_\infty = \|\mathcal{T}^* v^{(n-1)} - \mathcal{T}^* V^*\|_\infty \leq \gamma \|v^{(n-1)} - V^*\|_\infty.$$

Also, by the triangle inequality:

$$\|v^{(n-1)} - V^*\|_\infty \leq \|v^{(n-1)} - v^{(n)}\|_\infty + \|v^{(n)} - V^*\|_\infty.$$

Combining: $\|v^{(n)} - V^*\|_\infty \leq \gamma(\|v^{(n-1)} - v^{(n)}\|_\infty + \|v^{(n)} - V^*\|_\infty)$, so $(1-\gamma)\|v^{(n)} - V^*\|_\infty \leq \gamma\|v^{(n)} - v^{(n-1)}\|_\infty$, giving the result. □

Remark 4.4. The a posteriori bound provides a computable stopping criterion: terminate when $\|v^{(n)} - v^{(n-1)}\|_\infty < \varepsilon(1-\gamma)/\gamma$, which guarantees $\|v^{(n)} - V^*\|_\infty < \varepsilon$. This is the standard stopping rule used in practice.

5 Approximate Optimality of Greedy Policies

In practice, we terminate value iteration after finitely many steps, obtaining an approximate value function $\tilde{v} \approx V^*$. The greedy policy $d_{\tilde{v}}$ with respect to $\tilde{v}$ is then used as the agent's policy. How close is $V^{d_{\tilde{v}}}$ to V^*?

Theorem 5.1 (Approximate Policy Extraction)**.** *Let $\tilde{v} \in \mathbb{R}^S$ satisfy $\|\tilde{v} - V^*\|_\infty \leq \varepsilon$, and let $d_{\tilde{v}}$ be the greedy policy with respect to $\tilde{v}$. Then*

$$\|V^{d_{\tilde{v}}} - V^*\|_\infty \leq \frac{2\gamma\varepsilon}{1-\gamma}.$$

Proof. Since $d_{\tilde{v}}$ is greedy with respect to $\tilde{v}$, for every s, $(\mathcal{T}^{d_{\tilde{v}}}\tilde{v})(s) = (\mathcal{T}^*\tilde{v})(s)$.

Step 1: Show $\|\mathcal{T}^{d_{\tilde{v}}}\tilde{v} - V^{d_{\tilde{v}}}\|_\infty \leq \frac{\gamma\varepsilon}{1-\gamma}$.

Since $V^{d_{\tilde{v}}}$ is the fixed point of $\mathcal{T}^{d_{\tilde{v}}}$:

$$\begin{aligned}\|\mathcal{T}^{d_{\tilde{v}}}\tilde{v} - V^{d_{\tilde{v}}}\|_\infty &= \|\mathcal{T}^{d_{\tilde{v}}}\tilde{v} - \mathcal{T}^{d_{\tilde{v}}}V^{d_{\tilde{v}}}\|_\infty \\ &\leq \gamma\|\tilde{v} - V^{d_{\tilde{v}}}\|_\infty.\end{aligned}$$

By the triangle inequality: $\|\tilde{v} - V^{d_{\tilde{v}}}\|_\infty \leq \|\tilde{v} - V^*\|_\infty + \|V^* - V^{d_{\tilde{v}}}\|_\infty \leq \varepsilon + \|V^* - V^{d_{\tilde{v}}}\|_\infty$.

Step 2: Bound $\|V^* - V^{d_{\tilde{v}}}\|_\infty$.

Since $V^{d_{\tilde{v}}} \leq V^*$ componentwise (no policy beats the optimal), we have $\|V^* - V^{d_{\tilde{v}}}\|_\infty = \max_s[V^*(s) - V^{d_{\tilde{v}}}(s)]$.

For any state s:

$$\begin{aligned}V^*(s) - V^{d_{\tilde{v}}}(s) &= (\mathcal{T}^*V^*)(s) - V^{d_{\tilde{v}}}(s) \\ &\leq (\mathcal{T}^*\tilde{v})(s) + \gamma\varepsilon - V^{d_{\tilde{v}}}(s) \\ &= (\mathcal{T}^{d_{\tilde{v}}}\tilde{v})(s) + \gamma\varepsilon - V^{d_{\tilde{v}}}(s) \\ &= (\mathcal{T}^{d_{\tilde{v}}}\tilde{v})(s) - (\mathcal{T}^{d_{\tilde{v}}}V^{d_{\tilde{v}}})(s) + \gamma\varepsilon \\ &\leq \gamma\|\tilde{v} - V^{d_{\tilde{v}}}\|_\infty + \gamma\varepsilon.\end{aligned}$$

The first inequality uses $|(\mathcal{T}^*V^*)(s) - (\mathcal{T}^*\tilde{v})(s)| \leq \gamma\varepsilon$. The last inequality uses the contraction property.

Taking the maximum over s: $\|V^* - V^{d_{\tilde{v}}}\|_\infty \leq \gamma\|\tilde{v} - V^{d_{\tilde{v}}}\|_\infty + \gamma\varepsilon$.

Now $\|\tilde{v} - V^{d_{\tilde{v}}}\|_\infty \leq \varepsilon + \|V^* - V^{d_{\tilde{v}}}\|_\infty$, so

$$\|V^* - V^{d_{\tilde{v}}}\|_\infty \leq \gamma(\varepsilon + \|V^* - V^{d_{\tilde{v}}}\|_\infty) + \gamma\varepsilon = 2\gamma\varepsilon + \gamma\|V^* - V^{d_{\tilde{v}}}\|_\infty.$$

Rearranging: $(1-\gamma)\|V^* - V^{d_{\tilde{v}}}\|_\infty \leq 2\gamma\varepsilon$, giving the result. □

Near-Optimal Policy from Approximate VI

If value iteration produces $v^{(n)}$ with $\|v^{(n)} - V^*\|_\infty \le \varepsilon$, then the greedy policy $d^{(n)}$ satisfies

$$V^*(s) - V^{d^{(n)}}(s) \le \frac{2\gamma\varepsilon}{1-\gamma} \quad \text{for all } s \in \mathcal{S}.$$

Note the "error amplification" factor $2\gamma/(1-\gamma)$: an ε-accurate value function yields only an $O(\varepsilon/(1-\gamma))$-optimal policy. For $\gamma = 0.99$, this factor is approximately 198.

Error Amplification in Deep RL

The factor $2\gamma/(1-\gamma)$ in Theorem 5.1 explains a fundamental challenge in deep RL: function approximation error ε in the value network translates to policy suboptimality amplified by $1/(1-\gamma)$. This is why DQN and its variants are sensitive to the quality of the value function approximation, and why techniques like target networks and experience replay (which reduce ε) lead to significant practical improvements.

Example 5.2 (Value Iteration on a Two-State MDP)**.** Consider the two-state MDP from Lesson 39 with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a_1, a_2\}$, $\gamma = 0.5$:

s	a	$R(s,a)$	$P(s_1 \mid s,a)$	$P(s_2 \mid s,a)$
s_1	a_1	5	0.5	0.5
s_1	a_2	10	0	1
s_2	a_1	-1	0.8	0.2
s_2	a_2	2	0.3	0.7

We know from Lesson 41 that $V^*(s_1) \approx 13.04$ and $V^*(s_2) \approx 6.09$, achieved by $\pi^*(s_1) = a_2$, $\pi^*(s_2) = a_2$.

Starting from $v^{(0)} = (0,0)^\top$:

Iteration 1: $v^{(1)}(s_1) = \max\{5, 10\} = 10$, $v^{(1)}(s_2) = \max\{-1, 2\} = 2$. So $v^{(1)} = (10, 2)^\top$.

Iteration 2: $v^{(2)}(s_1) = \max\{5{+}0.5(0.5{\cdot}10{+}0.5{\cdot}2),\ 10{+}0.5(0{\cdot}10{+}1{\cdot}2)\} = \max\{8, 11\} = 11$. $v^{(2)}(s_2) = \max\{-1 + 0.5(0.8 \cdot 10 + 0.2 \cdot 2),\ 2 + 0.5(0.3 \cdot 10 + 0.7 \cdot 2)\} = \max\{3.2, 4.2\} = 4.2$. So $v^{(2)} = (11, 4.2)^\top$.

Iteration 3: $v^{(3)}(s_1) = \max\{5 + 0.5(0.5 \cdot 11 + 0.5 \cdot 4.2),\ 10 + 0.5 \cdot 4.2\} = \max\{8.80, 12.10\} = 12.10$. $v^{(3)}(s_2) = \max\{-1{+}0.5(0.8{\cdot}11{+}0.2{\cdot}4.2),\ 2{+}0.5(0.3{\cdot}11{+}0.7{\cdot}4.2)\} = \max\{3.82, 4.12\} = 5.12$.

Correcting: $v^{(3)}(s_2) = \max\{-1 + 0.5(8.8 + 0.84),\ 2 + 0.5(3.3 + 2.94)\} = \max\{3.82, 5.12\} = 5.12$.

So $v^{(3)} = (12.10, 5.12)^\top$.

The errors are $\|v^{(n)} - V^*\|_\infty$: after 1 iteration ≈ 4.09, after 2 ≈ 2.04, after 3 ≈ 0.97. The ratio is approximately $0.5 = \gamma$, confirming geometric convergence.

6 Value Iteration for Action Values

Value iteration can be stated directly for the action-value function, which is the version most relevant to model-free RL.

Definition 6.1 (Q-Value Iteration)**.** *Q-value iteration* generates a sequence $\{q^{(n)}\}_{n\ge 0}$ in $\mathbb{R}^{S\times A}$ by:

1. **Initialize:** $q^{(0)} \in \mathbb{R}^{S\times A}$ (commonly $q^{(0)} = 0$).

2. **Iterate:** For $n = 0, 1, 2, \ldots$,

$$q^{(n+1)}(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} q^{(n)}(s',a').$$

Proposition 6.2 (Convergence of Q-Value Iteration)**.** *The Q-value iteration sequence satisfies* $\|q^{(n)} - Q^*\|_\infty \le \gamma^n \|q^{(0)} - Q^*\|_\infty$*, and* $q^{(n)} \to Q^*$ *as* $n \to \infty$*.*

Proof. The operator $\mathcal{T}_Q^*$ defined by $(\mathcal{T}_Q^* q)(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} q(s',a')$ is a γ-contraction on $(\mathbb{R}^{S \times A}, \|\cdot\|_\infty)$ with fixed point Q^* (Lesson 41, Proposition 8.1). The Banach fixed-point theorem gives the geometric convergence bound. □

From Q-Value Iteration to Q-Learning

Q-value iteration requires the model (P, R) to compute the backup. Q-learning replaces the expected backup $\sum_{s'} P(s' \mid s,a) \max_{a'} q(s',a')$ with a single sampled transition: $q(s,a) \leftarrow (1-\alpha_t) q(s,a) + \alpha_t [r + \gamma \max_{a'} q(s',a')]$. Under appropriate step-size conditions ($\sum_t \alpha_t = \infty$, $\sum_t \alpha_t^2 < \infty$) and sufficient exploration, Q-learning converges to Q^* with probability 1 – a result proved via stochastic approximation theory (Lesson 19).

7 Policy Iteration

We now present policy iteration, which alternates between exactly solving for the value of a policy and improving it.

Definition 7.1 (Policy Iteration)**.** The *policy iteration* (PI) algorithm generates a sequence of deterministic stationary policies $\{d^{(k)}\}_{k \ge 0}$ by:

1. **Initialize:** Choose any deterministic stationary policy $d^{(0)}$.
2. **Policy evaluation:** Compute $V^{d^{(k)}}$ by solving the linear system $v = r^{d^{(k)}} + \gamma P^{d^{(k)}} v$, i.e., $V^{d^{(k)}} = (I - \gamma P^{d^{(k)}})^{-1} r^{d^{(k)}}$.
3. **Policy improvement:** Set $d^{(k+1)}$ to be the greedy policy with respect to $V^{d^{(k)}}$:

$$d^{(k+1)}(s) = \arg\max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^{d^{(k)}}(s') \right].$$

4. **Stopping:** If $d^{(k+1)} = d^{(k)}$, stop and output $d^{(k)}$.

Theorem 7.2 (Finite Convergence of Policy Iteration)**.** *Policy iteration converges to an optimal policy* π^* *in at most* $|\Pi^{\mathrm{SD}}| = A^S$ *iterations. Moreover, the sequence of values is monotonically non-decreasing:* $V^{d^{(k+1)}}(s) \ge V^{d^{(k)}}(s)$ *for all* s *and* k*.*

Proof. By the policy improvement theorem (Lesson 41, Theorem 7.2), $V^{d^{(k+1)}} \ge V^{d^{(k)}}$ with equality if and only if $V^{d^{(k)}} = V^*$. Since:

1. The value sequence is non-decreasing.
2. There are finitely many deterministic stationary policies (A^S), each with a distinct value function (if a value function repeats, the greedy policy must be the same, triggering the stopping condition).

The algorithm must terminate after at most A^S iterations. At termination, $d^{(k+1)} = d^{(k)}$, which means $V^{d^{(k)}}(s) = \max_a Q^{d^{(k)}}(s,a)$ for all s – the Bellman optimality equation. Thus $V^{d^{(k)}} = V^*$ and $d^{(k)} = \pi^*$. □

Example 7.3 (Policy Iteration on a Two-State MDP)**.** Using the same two-state MDP as Example 5.2:

Iteration 0: Start with $d^{(0)}(s_1) = a_1$, $d^{(0)}(s_2) = a_1$.

Policy evaluation: $V^{d^{(0)}} = (I - 0.5P^{d^{(0)}})^{-1} r^{d^{(0)}}$. With $r^{d^{(0)}} = (5, -1)^\top$ and $P^{d^{(0)}} = \left(\begin{smallmatrix} 0.5 & 0.5 \\ 0.8 & 0.2 \end{smallmatrix}\right)$, solving gives $V^{d^{(0)}} \approx (7.39, 2.17)^\top$.

Policy improvement: $Q^{d^{(0)}}(s_1, a_2) = 10 + 0.5 \cdot 2.17 = 11.09 > 7.39$, so $d^{(1)}(s_1) = a_2$. $Q^{d^{(0)}}(s_2, a_2) = 2 + 0.5(0.3 \cdot 7.39 + 0.7 \cdot 2.17) = 2 + 0.5(2.217 + 1.519) = 3.87 > 2.17$, so $d^{(1)}(s_2) = a_2$.

Iteration 1: $d^{(1)}(s_1) = a_2$, $d^{(1)}(s_2) = a_2$.

Policy evaluation gives $V^{d^{(1)}} \approx (13.04, 6.09)^\top$.

Policy improvement: checking all actions confirms $d^{(2)} = d^{(1)}$. Algorithm terminates: $\pi^* = d^{(1)}$.

Policy iteration found the optimal policy in just 2 iterations (1 improvement step), compared to the many iterations needed by value iteration.

Value Iteration vs. Policy Iteration

	Value Iteration	**Policy Iteration**
Per-iteration cost	$O(S^2A)$	$O(S^3 + S^2A)$
Convergence	Geometric, rate γ	Finite, $\leq A^S$ steps
Stopping criterion	$\|v^{(n)} - v^{(n-1)}\|_\infty < \epsilon$	$d^{(k+1)} = d^{(k)}$
Sensitivity to γ	Slow for $\gamma \approx 1$	Independent of γ

Policy iteration is typically preferred for moderate state spaces due to its fast convergence; value iteration scales better to large state spaces where exact policy evaluation (an $O(S^3)$ linear solve) is prohibitive.

8 Monotonicity of the Bellman Operator

The monotonicity property of $\mathcal{T}^*$ is used throughout the convergence analysis. We state and prove it formally.

Proposition 8.1 (Monotonicity of $\mathcal{T}^*$)**.** *The Bellman optimality operator is monotone: if $u(s) \leq v(s)$ for all $s \in \mathcal{S}$, then $(\mathcal{T}^*u)(s) \leq (\mathcal{T}^*v)(s)$ for all s. The same holds for $\mathcal{T}^\pi$ for any policy π.*

Proof. If $u \leq v$ componentwise, then for any s and a:

$$R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, u(s') \leq R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, v(s'),$$

since $P(s' \mid s,a) \geq 0$ and $u(s') \leq v(s')$. Taking the max over a on both sides preserves the inequality: $(\mathcal{T}^*u)(s) \leq (\mathcal{T}^*v)(s)$.

For $\mathcal{T}^\pi$, the argument is identical with the weighted sum $\sum_a \pi(a \mid s)[\cdots]$ replacing the max. □

Proposition 8.2 (Shift Property of $\mathcal{T}^*$)**.** *For any $v \in \mathbb{R}^S$ and constant $c \in \mathbb{R}$, define $(v+c)(s) = v(s) + c$. Then $\mathcal{T}^*(v + c) = \mathcal{T}^*v + \gamma c$.*

Proof. For any s:

$$\begin{aligned}(\mathcal{T}^*(v+c))(s) &= \max_a \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)(v(s')+c)\right] \\ &= \max_a \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)v(s') + \gamma c\right] \\ &= (\mathcal{T}^* v)(s) + \gamma c.\end{aligned}$$

□

Remark 8.3. The shift property implies that a uniform translation of the value function is "damped" by γ after one Bellman backup. Combined with monotonicity, this gives an alternative proof of the contraction property: if $\|u - v\|_\infty \le \delta$, then $v - \delta \le u \le v + \delta$, so by monotonicity $\mathcal{T}^*(v-\delta) \le \mathcal{T}^* u \le \mathcal{T}^*(v+\delta)$, and by the shift property $\mathcal{T}^* v - \gamma\delta \le \mathcal{T}^* u \le \mathcal{T}^* v + \gamma\delta$, giving $\|\mathcal{T}^* u - \mathcal{T}^* v\|_\infty \le \gamma\delta$.

This alternative characterization – contraction via monotonicity and shift – generalizes to settings where the explicit proof of contraction is more difficult.

9 Exercises

Exercise 9.1 (Value Iteration by Hand). (∗)
Consider the MDP with $\mathcal{S} = \{1,2,3\}$, $\mathcal{A} = \{L, R\}$, $\gamma = 0.9$:

s	a	$R(s,a)$	$P(1 \mid s,a)$	$P(2 \mid s,a)$	$P(3 \mid s,a)$
1	L	0	0.5	0.5	0
1	R	1	0	0	1
2	L	2	0.3	0.4	0.3
2	R	0	0	0	1
3	L	0	0	0	1
3	R	0	0	0	1

Starting from $v^{(0)} = (0,0,0)^\top$, compute $v^{(1)}$, $v^{(2)}$, and $v^{(3)}$. Verify that $\|v^{(n)} - v^{(n-1)}\|_\infty$ decreases by approximately a factor of $\gamma = 0.9$ at each step.

Exercise 9.2 (A Posteriori Stopping Criterion). (∗)
For the MDP in Exercise 5.2 with $\gamma = 0.5$, after 3 iterations we found $v^{(3)} = (12.10, 5.12)^\top$.

(a) Compute $\|v^{(3)} - v^{(2)}\|_\infty$.

(b) Use the a posteriori bound (Theorem 4.3) to bound $\|v^{(3)} - V^*\|_\infty$.

(c) Compare with the actual error $\|v^{(3)} - V^*\|_\infty$ (using $V^* \approx (13.04, 6.09)^\top$).

Exercise 9.3 (Policy Iteration Convergence). (∗∗)
Prove that during policy iteration, if $V^{d^{(k+1)}} = V^{d^{(k)}}$ at *any* state, then $V^{d^{(k+1)}} = V^{d^{(k)}}$ at *all* states and $d^{(k)}$ is optimal.

Hint: Use the Bellman optimality equation characterization.

Exercise 9.4 (Tightness of the Approximate Policy Bound). (∗∗)
Show that the bound $\|V^{d_{\tilde{v}}} - V^*\|_\infty \le 2\gamma\varepsilon/(1-\gamma)$ in Theorem 5.1 is tight up to a constant factor by constructing a two-state MDP where the policy loss is at least $\gamma\varepsilon/(1-\gamma)$ for an appropriately chosen $\tilde{v}$ with $\|\tilde{v} - V^*\|_\infty = \varepsilon$.

Hint: Use a deterministic MDP where the greedy action with respect to $\tilde{v}$ differs from the optimal action due to the approximation error.

Exercise 9.5 (Modified Policy Iteration)**.** $(***)$
Modified policy iteration performs only m steps of policy evaluation (rather than solving the linear system exactly): $v^{(k,0)} = v^{(k-1)}$, $v^{(k,j+1)} = \mathcal{T}^{d^{(k)}} v^{(k,j)}$ for $j = 0, \ldots, m-1$, then $v^{(k)} = v^{(k,m)}$ and $d^{(k+1)} = \text{greedy}(v^{(k)})$.

(a) Show that $m = 0$ gives value iteration and $m = \infty$ gives exact policy iteration.

(b) Prove that $\|v^{(k)} - V^*\|_\infty \leq \gamma^m \|v^{(k-1)} - V^*\|_\infty + (1-\gamma^m)\|V^{d^{(k)}} - V^*\|_\infty$.

(c) Show that modified policy iteration converges to V^* for any fixed $m \geq 1$.

10 Summary

- *Value iteration* computes $v^{(n+1)} = \mathcal{T}^* v^{(n)}$ and converges geometrically: $\|v^{(n)} - V^*\|_\infty \leq \gamma^n \|v^{(0)} - V^*\|_\infty$.
- The *a priori bound* starting from $v^{(0)} = 0$ gives $\|v^{(n)} - V^*\|_\infty \leq \gamma^n R_{\max}/(1-\gamma)$, requiring $O(\frac{1}{1-\gamma} \log \frac{R_{\max}}{\varepsilon(1-\gamma)})$ iterations for ε-accuracy.
- The *a posteriori bound* $\|v^{(n)} - V^*\|_\infty \leq \frac{\gamma}{1-\gamma}\|v^{(n)} - v^{(n-1)}\|_\infty$ provides a computable stopping criterion.
- A greedy policy from an ε-approximate value function is $\frac{2\gamma\varepsilon}{1-\gamma}$-optimal, demonstrating error amplification by $1/(1-\gamma)$.
- Q-value iteration applies the Bellman optimality operator for action values; it is the model-based template for Q-learning.
- *Policy iteration* alternates exact policy evaluation and greedy improvement, converging in finitely many steps ($\leq A^S$), independent of γ.
- The *monotonicity* and *shift properties* of $\mathcal{T}^*$ provide an alternative route to the contraction property and generalize to broader settings.
- Per iteration, VI costs $O(S^2 A)$ and PI costs $O(S^3 + S^2 A)$; VI suits large state spaces while PI suits small state spaces with $\gamma \approx 1$.

11 Further Reading

- **Lesson 43** develops refined error bounds for value iteration, stopping criteria, and proves monotonicity properties of VI sequences.
- **Lesson 44** extends the MDP framework to the average reward criterion, where the discount factor is absent and long-run average performance is optimized.
- Puterman [2014, Chapters 6–7] provides the definitive treatment of value iteration and policy iteration, including acceleration techniques and linear programming formulations.
- Bertsekas [2012, Chapter 2] extends the analysis to approximate dynamic programming, where Bellman backups use function approximation rather than exact tabular updates.
- Denardo [1967] places value iteration in the general framework of contraction mappings on ordered Banach spaces, yielding convergence results for a broad class of dynamic programming operators.
- Sutton and Barto [2018, Chapter 4] presents value iteration and policy iteration from the RL practitioner's perspective, with pseudocode and implementation details.

References

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